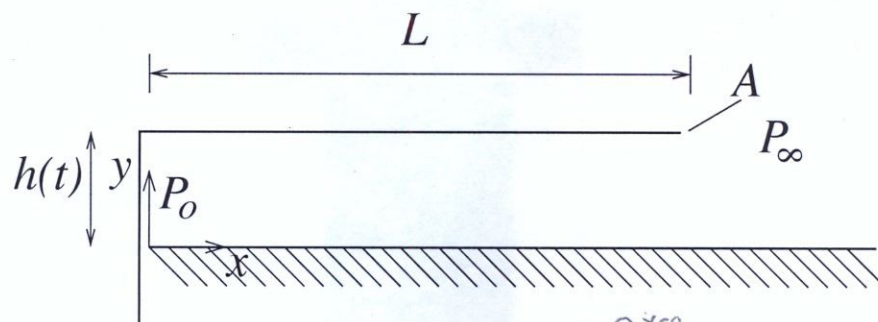


The L-shaped arm of length L depicted in the figure moves vertically approaching the horizontal wall, forming a thin channel of decreasing uniform thickness $h(t) \ll L$. A liquid of density ρ and viscosity μ fills the channel. Due to the arm displacement, the liquid is forced to move out, discharging to the surrounding atmosphere, where the reduced pressure is P_∞ . For a known value of $h(t)$,

1. Give the conditions for the motion in the channel to be dominated by viscosity.
2. Determine the velocity profile across the channel $v_x(x, y, t)$, the associated volume flux $Q = \int_0^h v_x dy$, and the reduced pressure distribution $P(x, t)$, including the value $P_0(t)$ at $x = 0$.
3. Calculate the force acting on the arm $\vec{F} = (F_x, F_y)$.
4. Obtain the torque exerted on the arm with respect to its end point A.



1) $\frac{\rho v_L h}{\mu} \frac{h}{L} \ll 1$, $v_L \sim \frac{L}{h} \dot{h} \Rightarrow \frac{h \dot{h}}{\nu} \ll 1$
 $\frac{h^2/\nu}{t_0} \ll 1 \Rightarrow \frac{h \dot{h}}{\nu} \ll 1$
 $t_0 \approx \frac{h}{\dot{h}}$

2) $0 = P_0 + \mu \frac{\partial^2 v_x}{\partial y^2} \Rightarrow v_x = \frac{P_0}{2\mu} y(h-y)$
 $Q = P_0 \frac{h^3}{12\mu} = -\frac{\partial P}{\partial x} \frac{h^3}{12\mu}$
 $\int_0^{h(t)} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial t} = 0 \right) dy \Rightarrow \frac{\partial Q}{\partial x} + \frac{dh}{dt} = 0 \Rightarrow -\frac{\partial}{\partial x} \left(P_0 \frac{h^3}{12\mu} \right) + \frac{dh}{dt} = 0$
 $-Q = \frac{\partial P}{\partial x} \frac{h^3}{12\mu} = \frac{dh}{dt} x + C_1 \Rightarrow P = \frac{12\mu}{h^3} \frac{dh}{dt} \frac{x^2}{2} + C_2 = \frac{6\mu}{h^3} \frac{dh}{dt} (x^2 - L^2) + P_\infty$
 $x=0, P = P_0 = -\frac{6\mu}{h^3} \frac{dh}{dt} L^2 + P_\infty$

3) $\vec{F}_y = \int_0^L (P - P_\infty) dx = -\frac{4\mu}{h^3} \frac{dh}{dt} L^3$
 $\vec{F}_x = -(P_0 - P_\infty) h - \int_0^L \mu \frac{\partial v_x}{\partial y} \bigg|_{y=h} dx = \frac{3\mu}{h^2} \frac{dh}{dt} L^2$
 $\frac{6\mu}{h^2} \frac{dh}{dt} L^2 - \frac{3\mu}{h^2} \frac{dh}{dt} L^2$

4) $\vec{M} = -\int_0^L (P - P_\infty)(L-x) dx \vec{e}_z = -\int_0^L \frac{6\mu}{h^3} \frac{dh}{dt} (x^2 - L^2)(L-x) dx \vec{e}_z = +\frac{5}{2} \frac{\mu}{h^3} \frac{dh}{dt} L^4 \vec{e}_z$